

V-I characteristics Measurement setup for Diodes



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Final Project

EE4130: PCB Design

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1 Brief Technical Proposal

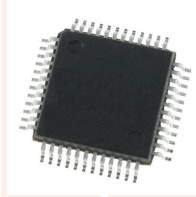
- This project proposes the design and implementation of a PCB-based I–V characteristics measurement system for semiconductor diodes using the STM32F303 microcontroller.
- The internal 12-bit DAC will generate a controlled voltage sweep, while the internal 12-bit ADC will measure the corresponding voltage across a sense resistor to compute current.
- A virtual ground technique will be employed to enable both forward and reverse bias characterization within a ± 1.65 V range.
- We will use the SWD interface in the STM32 to code and debug the microcontroller.
- The measured data will be transmitted to a host computer via UART for real-time visualization using a Python-based plotting interface.

The system aims to provide a compact, low-cost, and educational alternative to conventional laboratory instruments for diode characterization.

2 Circuit Overview

2.1 Core Microcontroller

The central processing unit of the proposed I–V characterization system is the STM32F303C8T6 microcontroller from STMicroelectronics. This device was selected due to its integrated mixed-signal architecture, combining high-performance digital processing with advanced analog peripherals. The microcontroller functions as the excitation controller, measurement unit, signal processor, and communication interface of the system.

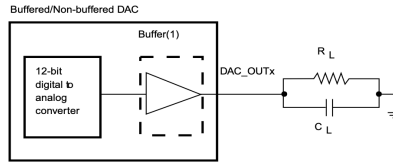


2.1.1 Digital-to-Analog Converter (DAC)

The STM32F303C8T6 includes an internal 12-bit voltage-output DAC. This DAC is used to generate the programmable excitation voltage required to sweep the diode under test. With a 3.3 V reference range, the DAC provides a resolution of approximately

$$\frac{3.3 \text{ V}}{4096} \approx 0.8 \text{ mV per step} \quad (0.1)$$

This fine voltage resolution enables controlled incremental sweeping of the diode across forward and reverse bias regions. The DAC output is digitally programmable, allowing automated voltage stepping. In the system, the DAC serves as a compact embedded voltage source.

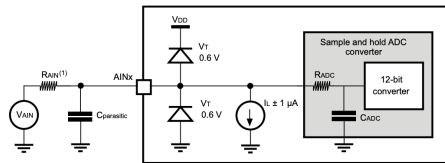


2.1.2 Analog-to-Digital Converter (ADC)

The microcontroller integrates multiple 12-bit ADC channels capable of sampling analog voltages with adequate precision. The ADC measures the voltage developed across the precision sense resistor placed in series with the diode. The current is computed digitally using

$$I = \frac{V_{\text{sense}}}{R_{\text{sense}}} \quad (0.2)$$

The 12-bit resolution provides approximately 0.8 mV per least significant bit, enabling detection of microampere-level currents when used with an appropriately selected sense resistor. The ADC also supports hardware averaging and oversampling, which improve effective resolution and reduce measurement noise.



2.1.3 UART Communication Interface

The built-in UART peripheral is used to transmit measured voltage and current data to a host computer. This enables real-time visualization and logging of the I-V curve using external software. The UART interface ensures reliable serial communication and seamless integration with USB-to-UART modules for PC connectivity.

2.1.4 Timers and Sweep Control

Internal timer peripherals are utilized to generate precise timing intervals for voltage stepping and sampling synchronization. Controlled timing ensures stable measurement acquisition and repeatable sweep behavior, which is critical for accurate device characterization.

2.1.5 Clock and Processing Core

The STM32F303C8T6 is based on a 32-bit ARM Cortex-M4 core, which provides sufficient computational capability to perform real-time data acquisition, arithmetic processing, and communication tasks simultaneously. The internal clock system ensures stable operation of the DAC, ADC, timers, and communication peripherals.

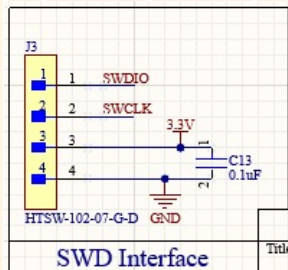
2.1.6 Role in the Overall System

Within the proposed measurement platform, the microcontroller performs the following key functions:

- Generation of programmable excitation voltage
- Acquisition of analog measurement signals
- Real-time current computation
- Control of sweep sequencing and timing
- Transmission of data to external visualization software

Thus, the STM32F303C8T6 serves as the central control and processing unit, integrating signal generation, measurement, computation, and communication within a single embedded platform.

2.1.7 SWD Interface for Coding and Debugging



This allows for easy communication with the microcontroller, and easier to debug.

2.1.8 Justification for Selection

The STM32F303C8T6 was selected due to its integrated analog peripherals, sufficient resolution for educational-level semiconductor characterization, reduced external component requirements, cost-effectiveness, and flexibility in configuration. Its mixed-signal capabilities make it particularly suitable for compact PCB-based measurement systems.

2.2 Voltage Generation Concept

The STM32 ADC cannot read negative voltages directly. Therefore, an op-amp level shifting circuit is used. A reference voltage of 1.65V (midpoint of 3.3V supply) is added to the input signal.

2.2.1 Op-Amp Based Negative Voltage Generation and Level Shifting

To apply both positive and negative input voltages to the device under test (DUT), an op-amp based signal conditioning circuit is used as shown in the given concept diagram. Since the STM32 ADC can only read voltages between 0V and 3.3V, negative voltages must be shifted into this measurable range.

2.2.2 Concept

The circuit uses:

- A reference voltage of 1.65 V (midpoint of 3.3 V supply)
- A DAC-generated signal from STM32
- Resistor network with an op-amp in summing configuration

The op-amp combines the DAC waveform and the 1.65 V reference to generate a level-shifted output voltage. This allows the system to represent negative input voltages in a positive voltage range that can be safely read by the ADC.

2.2.3 Working Principle

Let:

- $V_{ref} = 1.65 \text{ V}$ (mid reference)
- $V_{DAC} = \text{waveform generated from STM32 DAC}$

The op-amp circuit produces an output:

$$V_{out} = V_{DAC} - 1.65 \text{ V}$$

This means:

- When $V_{DAC} = 1.65 \text{ V}$, output is 0 V
- When $V_{DAC} > 1.65 \text{ V}$, output is positive
- When $V_{DAC} < 1.65 \text{ V}$, output represents negative voltage

Thus, the op-amp acts as a level shifter and differential amplifier, enabling both positive and negative voltage sweeps across the DUT while keeping ADC inputs within safe limits.

2.2.4 Purpose in This Project

This circuit is used to:

- Generate a bipolar sweep voltage for V-I testing
- Shift negative voltages into ADC measurable range

- Provide accurate and stable input excitation to the DUT

Hence, the op-amp based level shifting stage is an essential block for obtaining correct V-I characteristics using a unipolar ADC system.

2.3 USB Interface Design

The USB interface is kept separate from the STM32 using a CP2102 USB-to-UART converter circuit.

2.3.1 Working

- STM32 TX connects to CP2102 RX
- STM32 RX connects to CP2102 TX
- CP2102 converts UART data to USB signals
- PC reads data as a virtual COM port

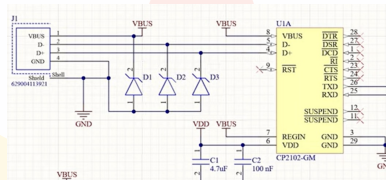


Fig. 0.1: Rough design schematic for reference

Thus, measured values from STM32 reach the PC through the CP2102 USB circuit.

3 PCB Architecture

The proposed PCB is organized into functional blocks consisting of the microcontroller unit, analog signal conditioning stage, diode test interface, power regulation section, and communication interface. The layout is designed to separate analog and digital domains to minimize noise and ensure stable measurements.

3.1 STM32F303C8T6 Microcontroller

The STM32F303C8T6 serves as the central control unit of the system. It generates the excitation voltage through its internal DAC, acquires measurement data using the internal ADC, controls sweep timing, and communicates measurement results to the host computer via UART.

3.2 Virtual Ground Network

A precision resistor divider is used to generate a mid-supply reference voltage (1.65 V), which is buffered using an operational amplifier. This virtual ground enables bipolar voltage sweeping without requiring dual power supplies.

3.3 Differential Amplifier (Subtractor Stage)

An operational amplifier configured as a differential amplifier subtracts the virtual ground from the DAC output. This stage produces a controlled $\pm 1.65\text{ V}$ excitation signal for both forward and reverse bias characterization.

3.4 Sense Resistor ($4.7\text{ k}\Omega$)

A precision $4.7\text{ k}\Omega$ resistor is placed in series with the diode under test. The voltage developed across this resistor is proportional to the diode current and is measured by the ADC to compute current using Ohm's law.

3.5 Diode Test Interface

A dedicated test terminal is provided for connecting the diode under characterization. This interface ensures secure electrical connection while allowing easy replacement of test devices.

3.6 Decoupling Capacitors

Bypass capacitors are placed near the microcontroller and operational amplifier supply pins to suppress noise and stabilize power rails. Proper decoupling improves measurement accuracy and system reliability.

3.7 Power Supply Section

The PCB includes a regulated 3.3 V supply to power the microcontroller and analog circuitry. Stable voltage regulation ensures consistent DAC and ADC performance.

3.8 UART Interface and USB-to-Serial Module

A UART header is provided for serial communication with a host computer through an external USB-to-Serial converter. This enables real-time data transmission for visualization and logging.

3.9 Clock and Reset Circuitry

An external crystal oscillator and associated load capacitors provide a stable clock source for precise timing control. A reset circuit ensures reliable system initialization during power-up.

4 Block Diagram

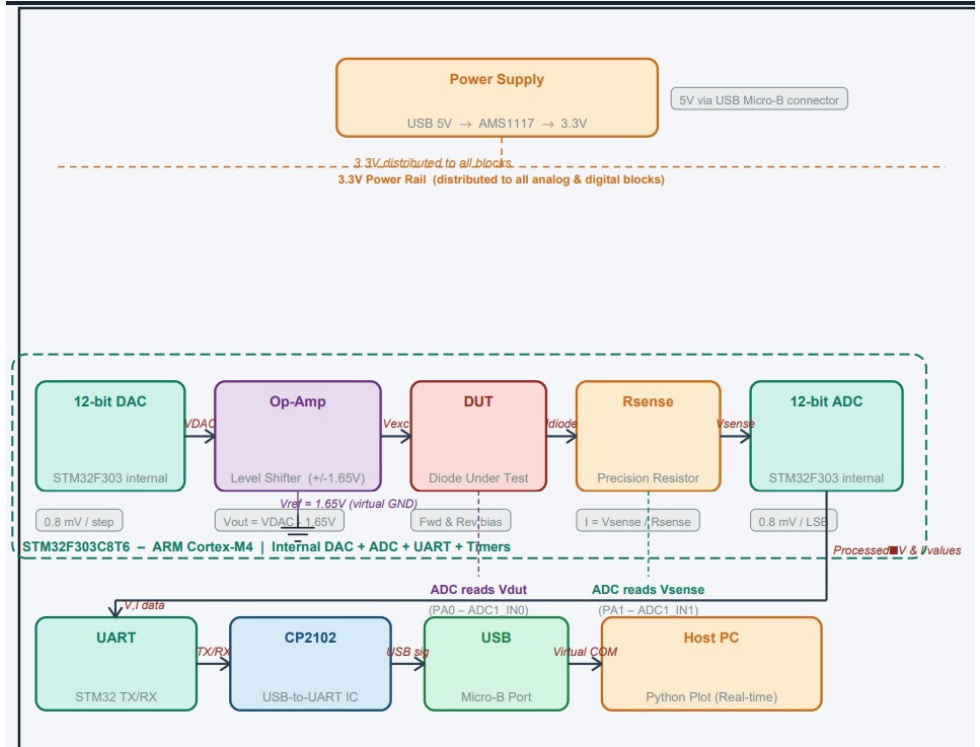


Fig. 0.2: Approximate Block Diagram and Project Flow

5 Estimated Cost Analysis (Bill of Materials)

The following table lists the major components required for fabrication and assembly of the proposed I-V characterization system. The cost estimation is approximate and based on typical academic procurement pricing.

Component	Qty	Approx. Cost (INR)
STM32F303C8T6 Microcontroller	1	600
Rail-to-Rail Operational Amplifier	2	100
Voltage Regulator (3.3V LDO)	1	100
Crystal Oscillator (8 MHz)	1	50
Load Capacitors (for crystal)	2	50
Precision Resistors (1%)	15	100
4.7 k Ω Precision Sense Resistor	1	50
Decoupling Capacitors (100nF, 10uF)	12	50
USB-to-UART Module (External)	1	200
UART Header Connector	1	50
Diode Test Terminal / Screw Terminal	1	100
Reset Push Button	1	50
Programming Header (SWD)	1	50
Miscellaneous Connectors and Hardware	-	500
Estimated Total		2050 INR (approx.)

The estimated total cost of the system is approximately 2000 -2500INR excluding the PCB cost.

In case external 16 bit DAC and 16 bit ADC are used instead of the inbuilt 12 bit, the cost might increase to 3000-4000, which improves precision and hence a clearer and accurate plot

6 Expected Outcome

- Accurate measurement of V-I characteristics
- Ability to measure both positive and negative voltage regions
- Real-time digital plotting on PC
- Compact and low-cost PCB-based measurement setup

7 Applications

- Diode and transistor characterization
- Analog electronics laboratory experiments